Architecture diagrams for HELIX

Six approaches, one shared architecture. Review the visual results before choosing a standard.

We are exploring how HELIX can produce clear solution diagrams for conference talks, developer onboarding, and reference documentation. The examples preserve the same six entities and relationships; differences in layout, typography, and interaction are deliberate.

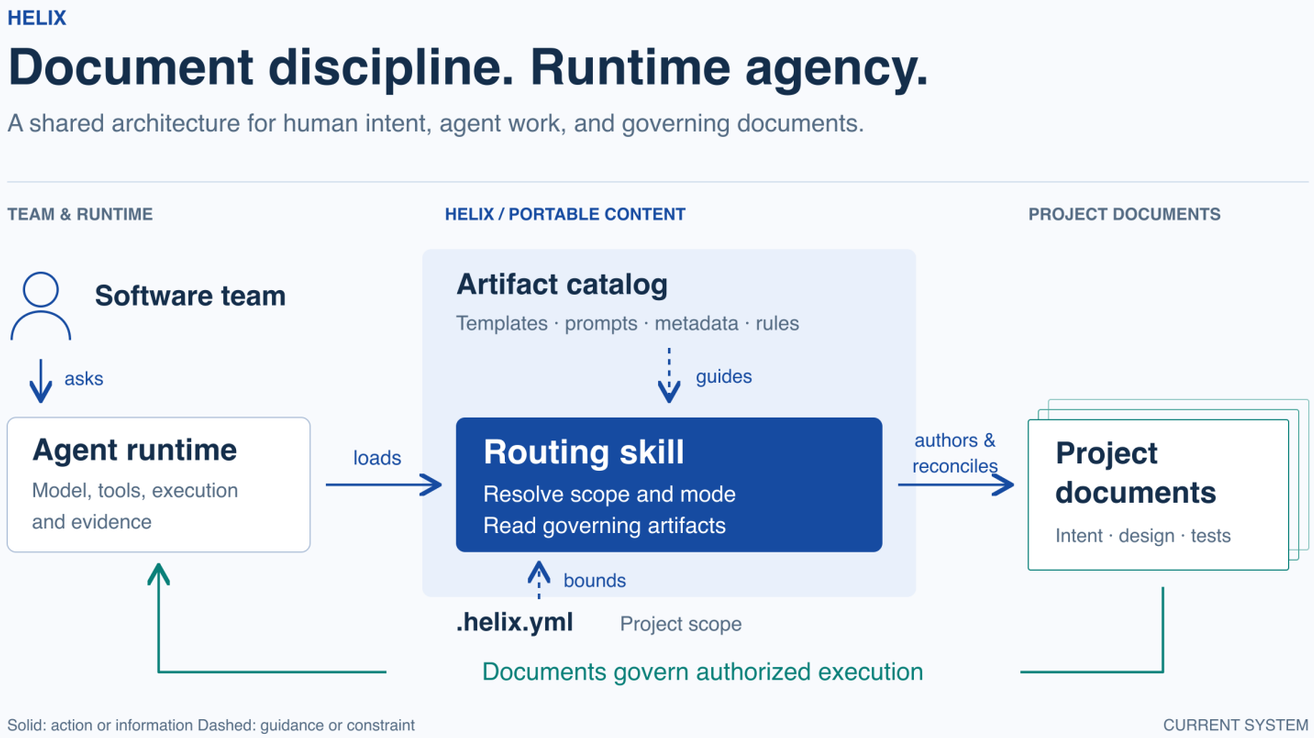
APPROACH	HOW IT WAS MADE	REVIEW FOCUS
1. Infographic Architect	Prompt-guided, hand-authored SVG	Composition and visible layout defects
2. Eraser Diagrams	Native components and connector routing	Clean geometry and editability
3. D2 / TALA	Declarative source with pinned positions	Layout control and revision effort
4. Visual Explainer	Prompt-guided HTML/SVG walkthrough	Visual style and explanatory focus
5. Beautiful Mermaid	Generated directly from Mermaid text	Maintainable documentation baseline
6. Integrated prototype	Eraser output + Visual Explainer style	Can the strengths work together?

Please judge the image first: what do you understand after a brief look, and what is hard to follow? Then consider how easily we could maintain it. The integration is a feasibility prototype, not an adopted HELIX standard.

Reading note: pages 2-7 show the six overview examples. Page 8 captures the integrated walkthrough in an active step. Page 9 lists sources and feedback questions. Static PDF captures cannot demonstrate interaction.

1. Infographic Architect

Original spike | Prompt-guided editable SVG



WHERE IT CAME FROM

Based on SizeSStudio's architecture-infographic skill, visual grammar, template catalog, and QA guidance. The three-column methodology pattern was adapted to HELIX. Geometry was authored locally; this was not an automatic layout engine.

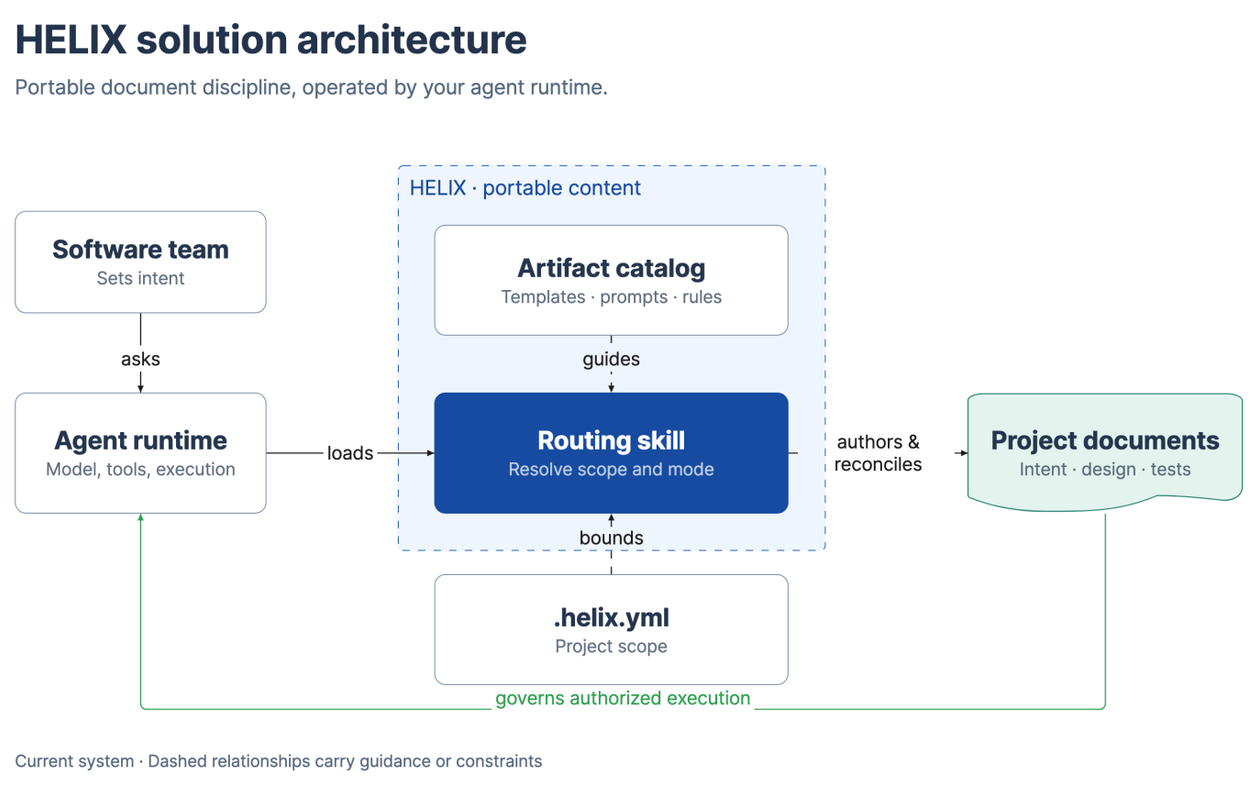
WHAT WE OBSERVED

The composition separates runtime, HELIX, and project documents. It also retains the crowded labels and ambiguous document-stack outlines identified in review. Those defects are shown rather than hidden in this comparison.

Source: github.com/SizeSStudio/infographic-architect | Source snapshot b3dd7e5

2. Eraser Diagrams

Native renderer | Components, measured text, and routed connectors



WHERE IT CAME FROM

Rendered with the open-source Eraser Diagrams Node package. The source is structured JSON with native groups, shapes, text, and relationships. Stock components were restyled with plain fills and an emphasized routing skill.

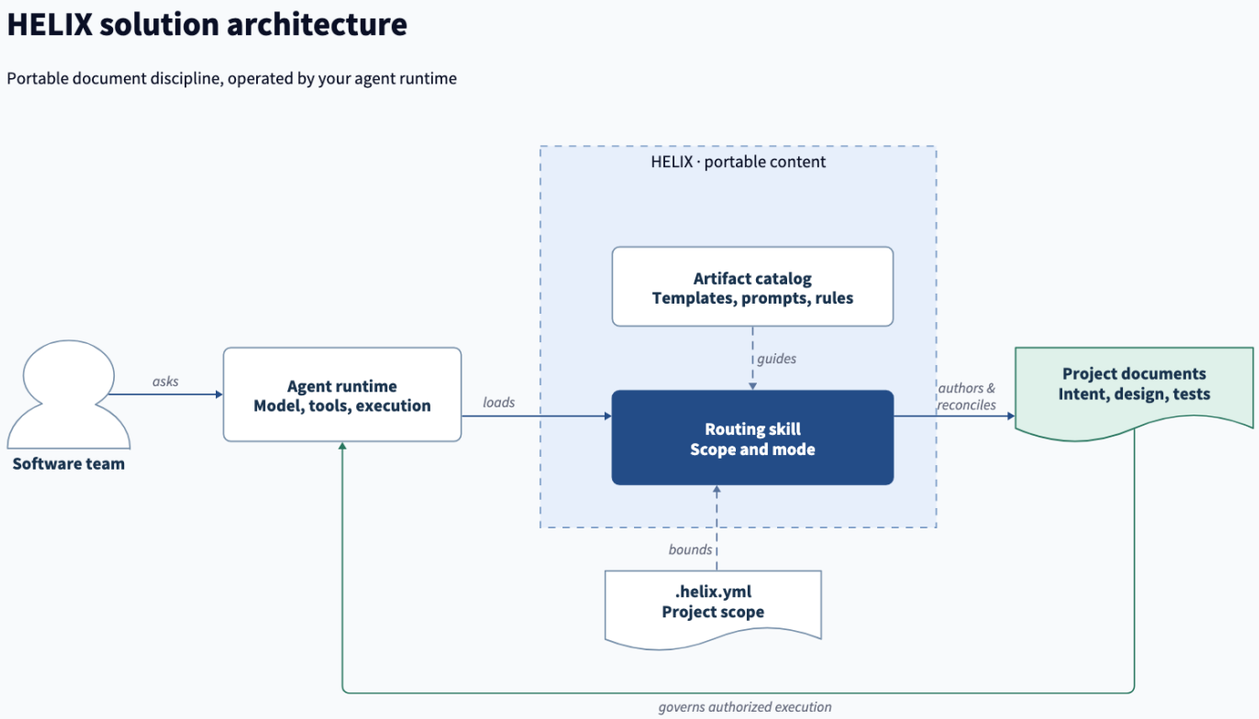
WHAT WE OBSERVED

The renderer measures component bounds and routes connections around them. This example returned no renderer warnings. It is cleaner geometrically, although the stock treatment needs more art direction for a conference presentation.

Source: github.com/eraserlabs/eraser-diagrams | @eraserlabs/diagrams 0.2.0

3. D2 / TALA

Declarative source | Positioned composition, TALA routing



WHERE IT CAME FROM

Produced with D2 0.9.0 and its TALA layout engine. An initial automatic layout was retained in the source bundle. The version shown pins entity positions while the engine handles rendering and connector routing.

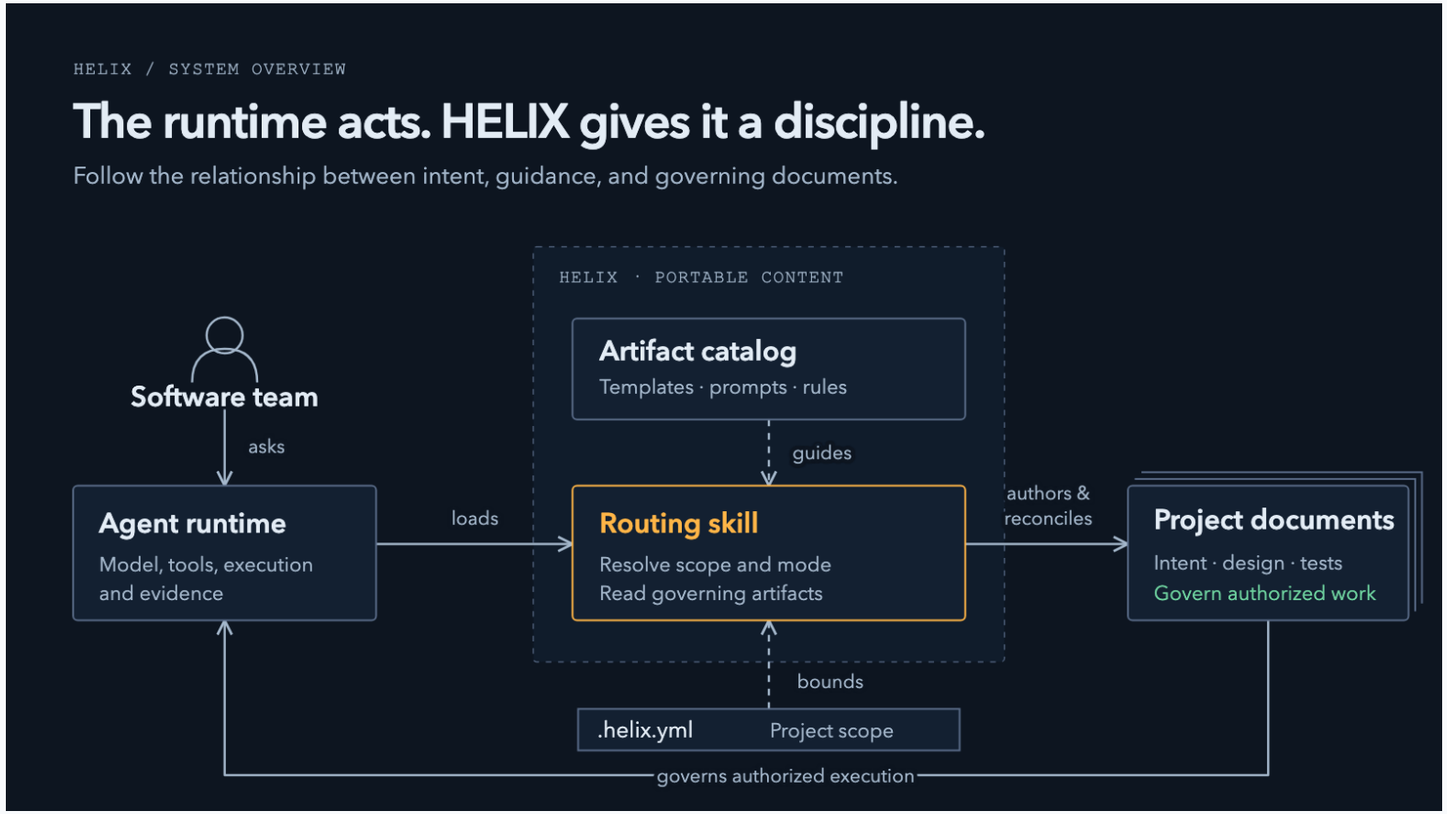
WHAT WE OBSERVED

The first automatic result was tall and poorly suited to a slide. Position constraints produced a usable landscape arrangement. This supports deliberate layout control; it does not show that automatic layout alone meets the presentation goal.

Sources: github.com/d2lang/d2 and d2lang.com/tour/tala/ | D2 0.9.0

4. Visual Explainer

Prompt-guided HTML/SVG | Dark Blueprint style and step-by-step focus



WHERE IT CAME FROM

Authored using nicobailon's Visual Explainer skill, Blueprint style register, and diagram guidance. The local HTML contains editable SVG and controls that highlight each of the six relationships. This was prompt-guided authoring.

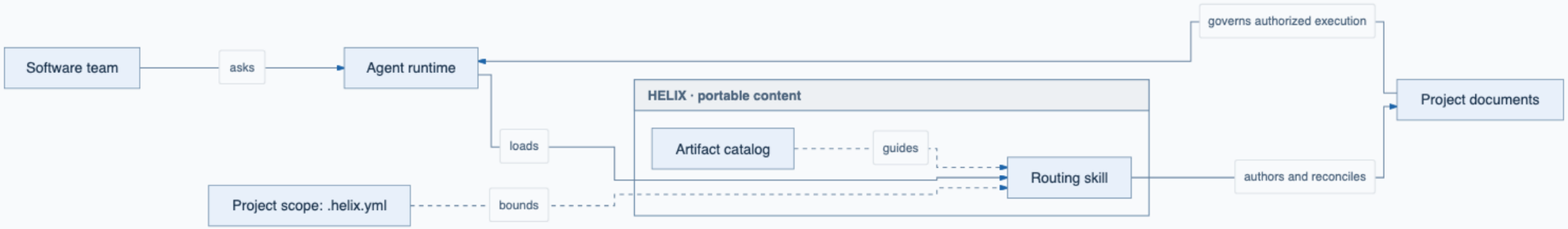
WHAT WE OBSERVED

The dark palette, warm focal accent, and active-path walkthrough were preferred in the initial review. The original figure still relied on manual geometry. This PDF shows the overview; the HTML also provides six-step interaction.

Source: github.com/nicobailon/visual-explainer | Source snapshot 69cd739

5. Beautiful Mermaid

Generated SVG | A low-effort documentation baseline



WHERE IT CAME FROM

Generated from a short Mermaid definition using the beautiful-mermaid package. The renderer provides coordinated colors and automatic layout. The skill and catalog were explicitly declared inside the HELIX subgraph.

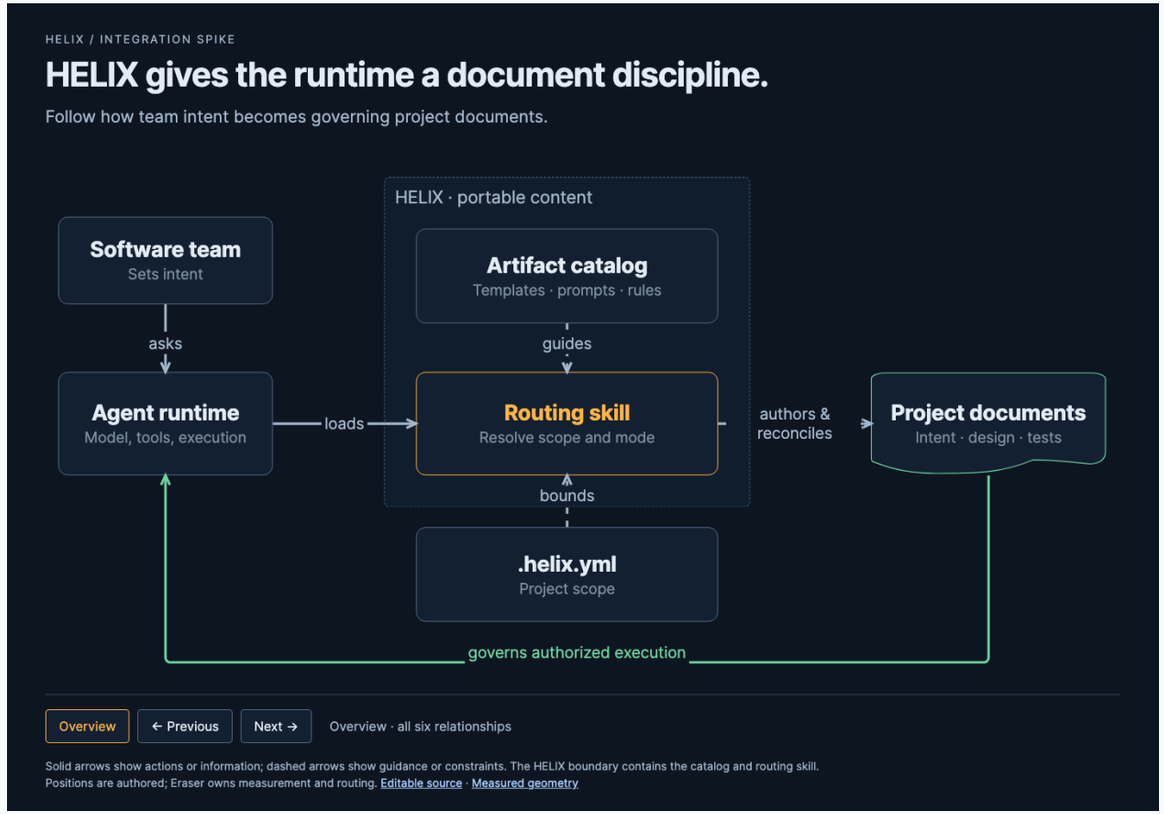
WHAT WE OBSERVED

Text-source maintenance is simple, but the result is very wide with thin connectors and relatively small labels. It is useful as a reference baseline. Its aspect ratio is retained here so the presentation tradeoff is visible.

Source: github.com/lukilabs/beautiful-mermaid | beautiful-mermaid 1.1.3

6. Integrated prototype

Visual Explainer presentation + Eraser native geometry



WHERE IT CAME FROM

The Eraser JSON was styled with Visual Explainer's dark Blueprint palette and rendered by Eraser. The native rendered elements, measured labels, and routed paths were retained. A local HTML wrapper adds captions and six-step controls.

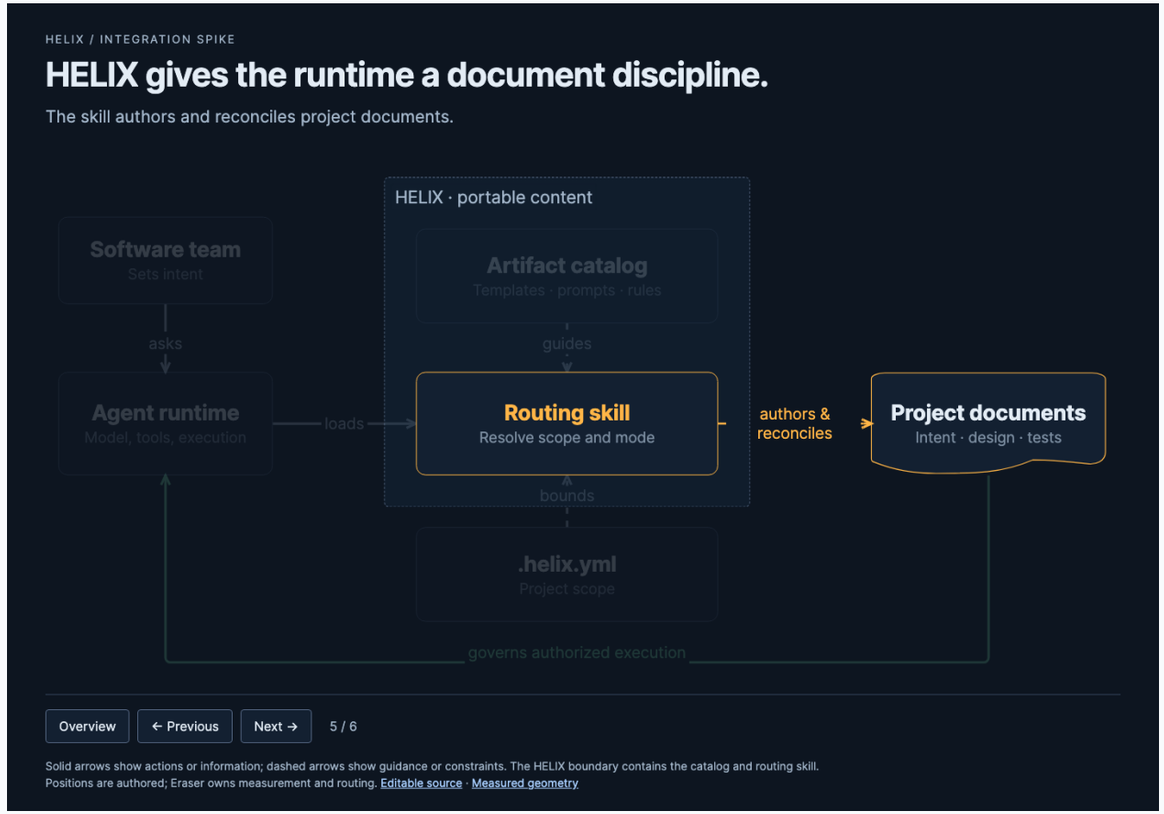
WHAT WE OBSERVED

The integration preserves geometry while changing emphasis. A governance return path needed an authored perimeter guide to avoid a connector crossing. Eraser still owns measurement and rendering; the walkthrough does not redraw the diagram.

Combined sources: Eraser Diagrams 0.2.0 + Visual Explainer guidance | Local feasibility prototype

Integrated walkthrough: an active relationship

Step 5 of 6 | Author and reconcile project documents



WHERE IT CAME FROM

This is a screenshot of the same integrated HTML at one active step. Controls select relationships by stable Eraser element IDs. Only opacity, color, border emphasis, and external captions change.

WHAT WE OBSERVED

All six steps and the Overview reset were exercised. Entity geometry remained unchanged and label-to-box overlap checks found none. The static PDF cannot reproduce these controls; the editable HTML remains the interactive prototype.

Evidence: local browser checks, measured Eraser geometry, and retained HTML source | No audience-comprehension or venue-projection test yet

Feedback and source references

Please review visual quality and maintenance practicality as separate questions.

READING

What is HELIX, what belongs inside it, and how does a request become a governing document?

GEOMETRY

Which labels or connectors are crowded, ambiguous, or difficult to trace?

PRESENTATION

Would you use the integrated dark diagram in a public talk? What would you change first?

ONBOARDING

Does stepping through one relationship at a time help, or would progressive reveal work better?

MAINTENANCE

Should HELIX favor editable native diagrams, concise text source, or different outputs for different audiences?

SOURCE PROJECTS

Infographic Architect

github.com/SizeSStudio/infographic-architect

Eraser Diagrams

github.com/eraserlabs/eraser-diagrams

D2 / TALA

d2lang.com/tour/tala/

Visual Explainer

github.com/nicobailon/visual-explainer

Beautiful Mermaid

github.com/lukilabs/beautiful-mermaid

C4 diagram review checklist

c4model.com/diagrams/checklist

The C4 checklist informed semantic review; it is not a renderer. These are conceptual solution views, not strict C4 container diagrams. Upstream projects did not create or endorse the HELIX examples.

Feedback format: name the example and page, identify a concrete strength or defect, and say whether it affects comprehension, visual polish, or future editing. The original SVG/JSON/HTML/D2/Mermaid files are retained with the spike notes.